Performance Across Complaint Styles and Lengths

1. Audit question and scope

Does the classifier perform differently on formal, detailed complaints and informal, short complaints? This audit reports overall and per-category performance, text-length differences and heuristic style-group differences. It is an exploratory linguistic performance audit, not a demographic fairness certification or a causal study.

2. Experimental design

The saved experiment retained 5,042 unique, eligible narratives from a 10,000-record CFPB API window. It used a stratified 70/15/15 split (3,529/756/757), seed 42. TF-IDF was fitted only on training data. Linear SVM C=1 was selected using validation macro-F1; final results below use the held-out test set. The model was not retrained for this report.

Overall metric	Held-out result
Accuracy	88.90% (673 correct / 757)
Macro-F1	0.6834
Weighted-F1	0.8842
Accuracy 95% bootstrap interval	86.66%-91.02%

Accuracy measures the fraction correctly classified. Macro-F1 gives equal weight to each category; weighted-F1 weights categories by support. The gap between macro-F1 and weighted-F1 is important because the test set is strongly imbalanced.

3. Category-level performance

Category	Test n	Recall	F1
Bank accounts	45	80.00%	0.7579
Credit cards	44	79.55%	0.7527
Credit reporting	540	95.74%	0.9460
Debt collection	74	71.62%	0.7465
Money services	14	64.29%	0.7200
Mortgages	15	80.00%	0.8889
Payday/title/personal loans	6	0.00%	0.0000
Student loans	9	66.67%	0.7500
Vehicle loans	10	50.00%	0.5882

Key concern: Credit reporting accounts for 540/757 cases (71.33%). Payday/title/personal loans has zero recall across six test cases. High overall accuracy therefore does not imply uniformly strong routing performance.

Length and Style Comparisons

4. Group definitions

Word counts are computed on original narrative text using the project's word-token rule. Short: 50 words or fewer; medium: 51-199; long: 200 or more. Length is observable, but is not a validated measure of formality or factual detail.

Length group	n	Accuracy	95% bootstrap interval
Short	118	83.90%	77.12%-89.83%
Medium	376	90.43%	87.23%-93.09%
Long	263	88.97%	85.17%-92.40%

Long-complaint accuracy exceeds short-complaint accuracy by **5.08 percentage points**. This is a descriptive association. Group intervals overlap; no hypothesis test or interval for the difference was calculated, so statistical significance is not established.

5. Heuristic formality comparison

The informal-short proxy requires 50 or fewer words plus at least one informal marker (u, ur, pls, plz, lol, wtf, gonna, wanna or idk) or repeated exclamation/question marks. The formal-detailed proxy requires 200 or more words, at least one formal marker (furthermore, therefore, pursuant, sincerely, regarding, request, documentation or attached), and no informal marker. All other texts are unassigned.

Style proxy	n	Accuracy	95% bootstrap interval
Formal-detailed	113	85.84%	77.88%-91.15%
Informal-short	5	80.00%	40.00%-100.00%
Unassigned	639	89.51%	87.01%-91.71%

The observed proxy-group gap is **5.84 percentage points**, but the informal-short group contains only five cases (four correct). One changed outcome moves its accuracy by 20 points. Further, 639/757 cases (84.41%) are unassigned. These results do **not** establish a reliable formality-related performance disparity.

6. Uncertainty methodology

Saved accuracy intervals use 1,000 within-group bootstrap resamples with replacement and the 2.5th/97.5th percentiles (seed 42). They describe uncertainty conditional on this fixed model and sample; they do not capture training variability, sampling-window bias or future distribution changes. The five-case interval is especially unstable.

Source files: outputs/audit.csv and src/project.py:style, audit, interval. Group definitions are rules written by the project, not human-assigned formality labels.

Controlled Comparison and Responsible Conclusions

7. Product-composition control

For each product class, the code samples equally many short and long test complaints, taking the smaller group size. A class is included only if each group has at least five cases. It then averages the within-class accuracy differences with equal class weight. This controls observed product composition among eligible classes, not all possible confounders.

Eligible class	n per group	Short accuracy	Long accuracy
Bank accounts	7	57.14%	71.43%
Credit cards	5	80.00%	100.00%
Credit reporting	81	95.06%	95.06%
Debt collection	16	62.50%	87.50%

Across these four classes, the equal-class average long-minus-short gap is **14.82 percentage points**. This single seeded comparison uses 109 cases per group, excludes five categories and has no reported uncertainty interval. It is exploratory, not an estimate for the entire database. For the formality proxies, **zero classes met the minimum-size rule**; a product-controlled style gap could not be calculated.

8. Bias risks and limitations

Selection: The source is the first API window under default ordering, not a random database sample; its recorded dates span 26 August-4 September 2024.
Coverage: Published narratives and categories with sufficient examples are selectively represented. Rare classes with fewer than 40 retained examples were excluded.
Measurement: Formality keywords can mislabel texts, and length does not prove factual detail. No completed human-rating result is available.
Generalization: A random split within a narrow window does not establish performance on later complaints, languages or institutions.
Fairness: Demographic attributes were not audited. Absence of demographic analysis is not evidence of fairness.

9. Recommended follow-up (not yet completed)

Have two independent readers annotate formality without seeing predictions, resolve disagreements, record factual detail and report inter-rater agreement. Increase informal-short and minority-category coverage; repeat class-controlled comparisons with uncertainty estimates. Evaluate on a later temporal holdout and retain human review for ambiguous or underperforming categories.

10. Audit conclusion

The model achieves useful aggregate performance, but performance is uneven across products and lower on short complaints in this sample. The formal-detailed versus informal-short comparison remains inconclusive because of sparse coverage and heuristic labels. The evidence supports an academic prototype with documented limitations, not autonomous deployment or a claim that the model is unbiased.

Evidence: outputs/test_metrics.json, classification_report.csv, audit.csv, controlled_audit.json, data_manifest.json; data/provenance.json; src/project.py. Dataset source recorded by project: CFPB Consumer Complaint Database (consumerfinance.gov). All numerical results describe the delivered experiment, not a new run.